

The Capital Asset Pricing Model in Practice: An Empirical Study of Apples's Market Sensitivity
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I. Introduction

Everyone tries to make stock predictions, whether it's the technical traders on wall street, the business developers for companies, or the economics student looking at trading volume during commercial breaks of the World Series. There are many different ways to calculate expected returns on assets. A popular approach is the Markowitz Mean-Variance Model (1952); however, this model requires calculating an extensive covariance matrix for each asset in the investment portfolio – a daunting and impractical task even with statistical software. Enter the Capital Asset Pricing Model (CAPM), a quick and intuitive way to predict the expected rate of return of an asset by comparing its performance to the wider stock market. If the S&P 500 falls, how predictive is that for stock X to fall? CAPM is useful for individual investment portfolio construction and for assessing the potential impacts of systematic (or market) risk.

II. Model Summary

• Model Used	Capital Asset Pricing Model (CAPM) estimated with OLS regression
• Data Period	Oct 1, 2020 – Oct 1, 2025
• Assets	Apple Stock (AAPL) and S&P 500 Index (daily log returns)
• Risk-Free Rate	4.5% annualized (converted to daily rate)
• Beta (β)	1.26; Apple's returns move 26% more than the market
• R²	0.57; 57% of Apple's excess returns explained by market movement
• Predicted Return (CAPM)	11.12% annualized
• Actual Return (Observed)	11.92% annualized
• Abnormal Return	+0.80%; Apple slightly outperformed CAPM expectations

III. Data and Methodology

A. Data organization

The data collected was the daily closing price for Apple and the S&P 500 Index from October 1, 2020 to October 1, 2025, which was obtained from [nasdaq.com](https://www.nasdaq.com). Each daily price was converted to a daily log return:

$$R_t = \ln\left(\frac{P_t}{P_{t-1}}\right)$$

Taking the log difference helps to normalize the scale of the variables. While shares of Apple are represented in dollars, stock indexes are not. Taking the log transforms price changes into proportional (percentage) changes, making comparisons across assets easier.

The risk-free rate R_F is the *guaranteed* return, typically proxied through US Treasury Bills. If an investor sought constant, predictable returns and had a very low risk tolerance, their investment portfolio would mostly include treasury bills. The daily risk-free rate $R_{F,daily}$ was obtained by converting an estimated annual rate of 4.5%. We turn this into a daily value because we are tracking the daily changes in the assets's returns.

$$R_{F,daily} = \ln((1 + 0.045)^{1/252})$$

Throughout this analysis, I annualize variables using the value 252. This number is used to account for trading days per year. While each year will have different trading days, the industry standard is to approximate this as 252 days.

An important consideration for investors is the expected returns from a risky asset versus the risk-free rate, which again would be the return from a treasury bond. The difference between these returns is the *Excess Return*, calculated as

$$R_{Apple,Excess} = r_{Apple} - r_f \quad \text{and} \quad R_{S\&P\ 500,Excess} = r_{S\&P\ 500} - r_f$$

B. Estimating the model

The Capital Asset Pricing Model utilizes OLS Regression to estimate the expected return for Apple,

$$r_{Apple} - r_F = \alpha + \beta_{Apple} \times (r_{S\&P\ 500} - r_f)$$

where:

- r_{Apple} : Apple's daily return
- α : Apple's abnormal return not explained by the market (intercept)
- β_i : sensitivity of the stock to the overall market, or here, the market's excess returns.

The beta coefficient measures how sensitive Apple's returns are to changes in the overall market — a β above 1 means Apple tends to move more than the market in either direction. OLS regression is commonly used in economics because its coefficients provide clear, quantitative interpretations of relationships between variables. A β of 1.0 means that Apple tends to move in the same direction as the market, with the same magnitude. So, if the S&P 500 rises by 1 percent, Apple's return also increases by 1 percent. In other words, the Beta coefficient captures Apple's exposure to *systematic risk*, or the risk of the overall market.

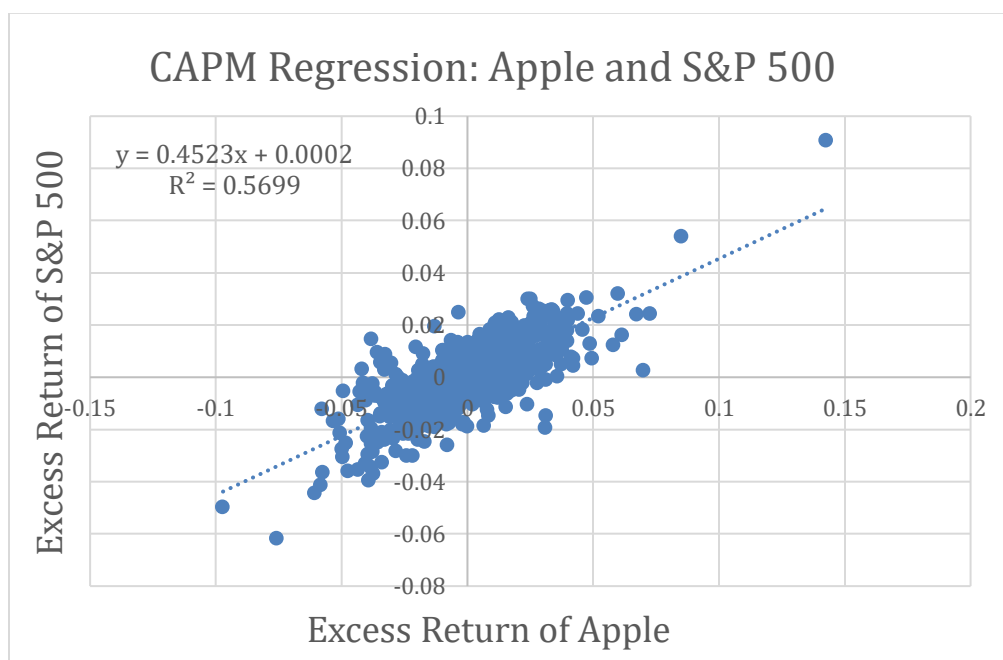
IV. Results:

The results from the regression analysis are reported in the following table.

Coefficient	Estimate	t-Statistic
α (alpha)	-0.00001848 (0.0003336)	-0.0554202
β (beta)	1.2598 (0.03093)	40.73
R^2	0.5699	

Our variable of interest is β , which measures Apple's systematic risk and sensitivity to overall market movements. A $\beta > 1$ indicates a stock that is more volatile than the market. Here, β has a value of 1.2598, meaning that if the S&P rises by 1%, Apple's return tends to rise by 1.25% (in the equation, $\beta \times 1\%$). This makes sense for a growth-oriented and profitable tech firm. This term is statistically significant (t-Statistic of 40.73), meaning there is a real relationship between the S&P and Apple's excess returns. The alpha term is statistically insignificant, suggesting there is no evidence of abnormal returns – the CAPM model predicts Apple's returns fairly accurately.

The R-squared value shows how well the regression line fits the data. In context, the R^2 value of 0.5699 indicates that 57% of Apple's daily excess returns are explained by market movements, or the model explains 57% of Apple's day-to-day return movements, the rest comes from unaccounted for variables. That is strong for a single factor (single variable) regression model. Below is a regression graph, illustrating the excess returns of each asset and the regression line estimated through the model.



After completing the CAPM regression, it is necessary to compare the model expected Apple return versus what was observed. This validates that the results for alpha and beta align with the realized performance. First, I calculated the mean daily return from the daily log returns of Apple over the period. I then annualized this value by multiplying it by 252 (the number of trading days) to obtain a continuously compounded log return. This value represents a continuously compounded (log) return, whereas the CAPM variables are based on simple returns. The following operation was performed to make the annualized value comparable to the CAPM model:

$$R_{Annual} = e^{R_{Annual,log}} - 1$$

The realized annualized return for Apple was 11.92% while the CAPM model predicted 11.12%. The difference between these two, the Abnormal Return, is 0.8004%, showing that Apple slightly outperformed the return predicted by CAPM. Overall, it seems CAPM does predict the value of Apple fairly well.

V. Implication

This analysis has implications in two important scenarios. First, if an investor wanted to increase their holdings of risky assets, chasing the higher returns from investing only in bonds (under the risk-free rate) or the index funds, they could compute CAPM models for a variety of stocks. These results show that, on average, Apple moves above the S&P by 25%. Other stocks may exhibit even greater sensitivity. CAPM's original purpose was to help investors diversify their portfolios by balancing expected returns with risk exposure.

A second implication, and one that is more interesting, is CAPM's use in event study. Imagine on October 2, 2025 a whistleblower revealed damaging information about Apple and the stock plummeted 15%. Months later, after investigations proved the claims baseless, Apple could

use CAPM to estimate damages in a defamation lawsuit. If the S&P 500 had since risen while Apple's stock remained flat, CAPM would help isolate how much of Apple's decline was abnormal and attributable to the false report rather than overall market conditions.

VI. Criticism of CAPM

While CAPM is useful, especially because of its ease of calculations and interpretability, it is not a perfect model. Most of the criticism comes from the underlying assumptions that generally are violated, specifically that all individuals have a 1-period planning horizon (this means investors do not hold long-term positions) and have identical expectations about the markets and individual assets. It is also a simplified model, relying only on the covariance between a stock and the overall market to capture risk. Richer financial models may yield more accurate predictions, such as the work by Fama–French (1993) or Carhart (1997).

Furthermore, OLS regression assumes that residuals are independent over time, or that there is no autocorrelation. In his work on the Efficient Market Hypothesis, Fama (1969) emphasizes that stocks should follow a “random walk,” meaning a today's return should not depend on yesterday's. In reality, investor momentum, news cycles, and seasonal trends do occur. From a strict econometric standpoint, this suggests OLS is not an acceptable tool for this analysis.